AdifySkills

n8n Automation Mastery
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MENTOR COPY

Module 6 · Production-Grade Engineering

DAY 19 of 25

# Self-Hosting n8n: Docker, VPS & Domains

"Apna khud ka n8n server — unlimited workflows."

⚡ AAJ HUM YEH BANAYENGE

## Self-Hosted n8n + Uptime Monitor

Aaj VPS par Docker Compose se n8n + Postgres + Caddy (auto HTTPS) chalate hain — ready files neeche hain. Saath mein ek chhota workflow jo har 5 minute aapke n8n ka /healthz check karta hai aur down hone par Telegram alert deta hai.

Docker | Docker Compose | Postgres | Nginx/Caddy

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Self-hosting = unlimited executions, data apne paas, client ko kam cost. Har serious automator ko aana chahiye.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 Cloud vs self-hosted: cost & control**

🌍 Kiraya vs EMI.
⚙️ Cloud: easy, limits. Self-host: sasta, full control, maintenance.

**02 Docker & Docker Compose basics**

🌍 Tiffin box — sab ek dabbe mein.
⚙️ docker-compose.yml: n8n + postgres services, volumes.

**03 VPS setup (Hostinger VPS / any Ubuntu server)**

🌍 Khaali plot.
⚙️ Ubuntu VPS → SSH → Docker install → firewall.

**04 Custom domain + SSL (reverse proxy)**

🌍 Ghar ka naam board + taala.
⚙️ DNS A record → Caddy/Nginx reverse proxy → Let's Encrypt SSL.

**05 Environment variables & Postgres backend**

🌍 Ghar ki settings ek list mein.
⚙️ N8N_HOST, WEBHOOK_URL, N8N_ENCRYPTION_KEY, DB_TYPE=postgresdb.

**06 Updates, backups & restore**

🌍 Ghar ki duplicate chaabi.
⚙️ pg_dump nightly + encryption key safe; docker compose pull && up -d.

## ⏱️ Class ka plan (2 ghante)

| 10m | 30m | 55m | 15m | 10m |
|---|---|---|---|---|

`0:00-0:10` **Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

👩‍🏫 2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

`0:10-0:40` **Concept: Hosting Architecture**

🪄 *Kiraye ke ghar mein maalik ke rules. Apne ghar mein tumhare rules — par safai bhi tumhe karni hai.*

👩‍🏫 Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

`0:40-1:35` **Live Setup**

Fresh VPS → Docker → n8n + Postgres → domain + SSL

🪄 *Aaj ki kahani yaad hai — 'Kiraye ka flat vs apna ghar'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

👩‍🏫 Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

`1:35-1:50` **Checklist Review**

Security hardening

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

👩‍🏫 Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

1:50–2:00 **Q&A + Homework**

*🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

👩‍🏫 Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

## 🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### VPS

Cloud mein apna virtual computer (Ubuntu).

🪄 **Analogy:** Kiraye ka flat jahan aap kuch bhi rakh sakte ho

CONCEPT 02

### Docker

App + uski saari dependencies ek "container" mein.

🪄 **Analogy:** Tiffin box — sab packed

CONCEPT 03

### Reverse proxy + SSL

Domain → n8n, HTTPS ke saath (Caddy/Nginx).

🪄 **Analogy:** Building ka reception

CONCEPT 04

### Encryption key

Credentials encrypt karne ki key — kho gayi to credentials gaye.

🪄 **Analogy:** Locker ki master key

### 🔑 Aaj ka main idea

Self-hosting = n8n ko apne server (VPS) par Docker se chalana. Domain + HTTPS lagao, Postgres database use karo, aur regular backup lo.

- Hostinger VPS par n8n + Postgres
- n8n.meracompany.com HTTPS ke saath
- Har raat database backup Drive par

### 📖 Ek kahani se samjho: Kiraye ka flat vs apna ghar

1 🏢 **Cloud = kiraye ka flat** = n8n Cloud

2 🏠 **Apna plot liya** = VPS server

3 📦 **Ready-made ghar ka dabba** = Docker container

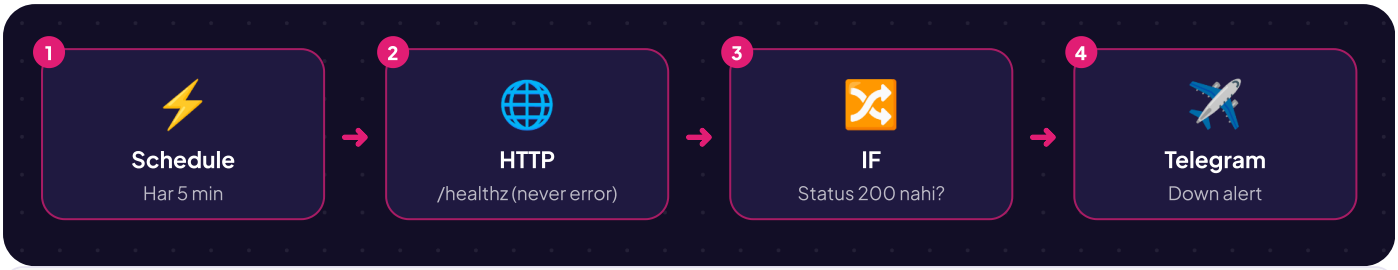
4 🪧 **Ghar ka address board** = Domain + SSL

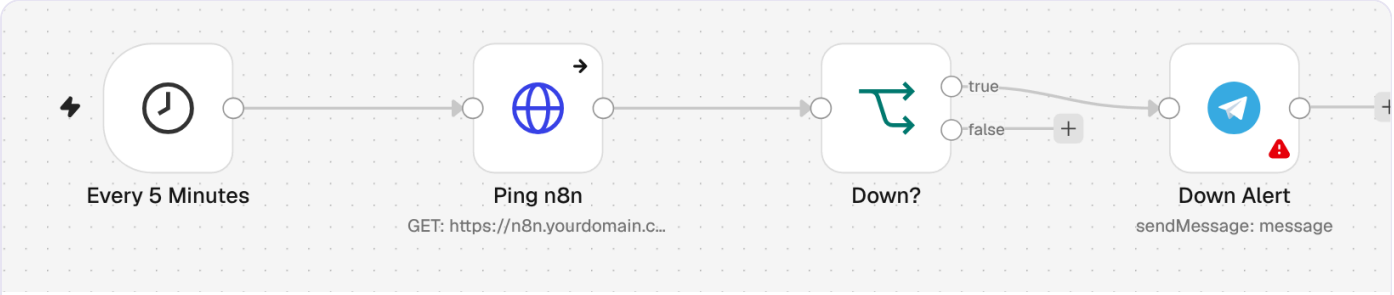
5 🗄️ **Tijori aur backup chaabi** = Postgres + backups

🪄 Ek line mein: “Kiraye ke ghar mein maalik ke rules. Apne ghar mein tumhare rules — par safai bhi tumhe karni hai.”

# 🗺️ Aaj ka workflow: Self-Hosted n8n + Uptime Monitor

Data left se right chalta hai. Har dabba ek n8n node hai.




📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

### 💰 Isse kitna kama sakte ho?

Clients jo apna n8n chahte hain (VPS setup + monitoring) — aam taur par **₹8,000 – ₹20,000 + maintenance**.
*(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

1. 🖥️ **SSH**
   VPS mein login, apt update
2. 🖥️ **SSH**
   Docker install: curl -fsSL https://get.docker.com | sh
3. 📄 **Read/Write Files from Disk**
   docker-compose.yml: n8n + postgres
4. 🖥️ **SSH**
   DNS A record → VPS IP; Caddy reverse proxy
5. 🖥️ **SSH**
   docker compose up -d
6. ⚡ **Schedule Trigger**
   Nightly backup workflow

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

### 🔑 Setup (import ke baad)

- [ ] Connect: **Telegram bot token (BotFather se)**
- [ ] URL mein apna domain daalo
- [ ] Best practice: yeh monitor kisi DOOSRE n8n / server par chalao (same server down hua to alert kaun bhejega?)

### ✅ Test kaise karein

1. Galat URL daal ke test karo → alert aana chahiye

## ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

**😩 Pehle (manual)**
Cloud plan ₹2000+/mahina, limits

➜

**⚡ Ab (n8n)**
VPS ~₹500/mahina, unlimited workflows

⏱ Kharcha kam, control poora — lekin maintenance tumhari.

## ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** — 📋 Select karke copy karo

```
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    environment:
      - N8N_HOST=n8n.yourbrand.com
      - WEBHOOK_URL=https://n8n.yourbrand.com/
      - DB_TYPE=postgresdb
      - N8N_ENCRYPTION_KEY=${N8N_ENCRYPTION_KEY}
    volumes: [n8n_data:/home/node/.n8n]
```

## 📦 Sample data — input se output tak

**📥 Input** — 📋 Select karke copy karo

```
{
  "domain": "n8n.yourbrand.com",
  "vps": "2 vCPU / 4 GB RAM"
}
```

**📤 Output** — 📋 Select karke copy karo

```
{
  "url": "https://n8n.yourbrand.com",
  "ssl": "valid",
  "db": "postgres"
}
```

# 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

**💬 Build it with ChatGPT / Claude** — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Aaj VPS par Docker Compose se n8n + Postgres + Caddy (auto HTTPS) chalate hain — ready files neeche hain. Saath mein ek chhota workflow jo har 5 minute aapke n8n ka /healthz check karta hai aur down hone par Telegram alert deta hai.

Flow: Schedule (Har 5 min) → HTTP (/healthz (never error)) → IF (Status 200 nahi?) → Telegram (Down alert)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

**💬 Debug helper prompt** — 📋 Select karke copy karo

```
I'm building the n8n workflow "Self-Hosted n8n + Uptime Monitor" (nodes: Every 5 Minutes, Ping n8n, Down?, Down Alert).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

**⚡ Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-19-uptime-monitor.json`

**🌐 Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/19`

# 🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo. |
| **Time par apne aap nahi chala?** | "Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo. |

### ⚠️ Yeh galtiyan sabse common hain

! WEBHOOK_URL set na karna — webhooks localhost dikhate hain
! Encryption key backup na karna
! Volume mount bhool jaana — restart par sab data gayab

### ⭐ Pro tips — time bachane wali tricks

★ Hostinger VPS par n8n ka 1-click template bhi milta hai
★ Updates se pehle backup + release notes padho
★ Firewall: sirf 22, 80, 443 open

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

## 🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

### 🛠 Live lab

Live: ek fresh Ubuntu VPS par n8n + Postgres Docker Compose ke saath, apne domain par HTTPS.

### 📝 Homework

Apna self-hosted instance setup karo aur ek workflow export/import karke migrate karo.

☐ Homework ho gaya

**BEGINNER**
Local Docker par n8n chalao ☐

**INTERMEDIATE**
VPS + domain + SSL ☐

**ADVANCED**
Automatic backups to S3/Drive ☐

### 🧩 Aaj ka challenge — pehle khud try karo

**🎯 Monitor ko 2 baar lagataar fail hone par hi alert bhejne do (false alarm kam).**

**💡 Hint 1**
Ek baar fail hone par thoda ruk ke dobara check karo.

**💡 Hint 2**
Down? true → Wait 1 minute → Ping dobara → IF phir bhi down → alert.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

# 🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. Webhooks ka public URL batane wala env?**

- (A) NODE_ENV
- (B) WEBHOOK_URL
- (C) PORT
- (D) TZ

**Q2. Credentials encrypt karne wali key?**

- (A) SSH key
- (B) SSL cert
- (C) N8N_ENCRYPTION_KEY
- (D) API_KEY

**Q3. Data restart ke baad bhi rahe — kya chahiye?**

- (A) Cache
- (B) Cookie
- (C) Nothing
- (D) Docker volume

**Q4. HTTPS ke liye?**

- (A) Reverse proxy + SSL cert
- (B) Bada server
- (C) Postgres
- (D) Redis

## 📖 Aaj ke naye shabd

**Workflow**
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

**Trigger**
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

**Docker**
Software ko ek ready dabbe (container) mein chalane ka tool — har jagah same chalta hai.

**VPS**
Kiraye ka online computer/server jo 24×7 chalta hai.

**SSL**
Website ka taala — https wala lock, data safe rehta hai.

**Schedule**
Fixed time par workflow chalana — roz, har ghante, har Monday.

# 👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] Fresh VPS par live setup — 30 min mein HTTPS par n8n
- [ ] Har step ke baad checkpoint: "sabka yahan tak hua?"
- [ ] Security hardening checklist share karo

## 🗣️ Har block ke liye mentor tip

**10m — Recap**

2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

Class se poochho: Kal wale workflow mein sabse mushkil step kaunsa laga? · Homework mein kahan error aaya aur kaise theek kiya? · Kal ka ek node naam batao aur woh kya karta hai?

**30m — Concept: Hosting Architecture**

Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

Class se poochho: Docker Compose kyun use karte hain? · WEBHOOK_URL env variable kyun zaroori? · Update se pehle kya karein?

**55m — Live Setup**

Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

Class se poochho: Is node ka output agle node mein kaise ja raha hai? · Agar yeh node fail ho to kya hoga? · Isko apne kaam/business mein kahan use kar sakte ho?

**15m — Checklist Review**

Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

Class se poochho: Webhooks ka public URL batane wala env? · Credentials encrypt karne wali key? · Data restart ke baad bhi rahe — kya chahiye?

**10m — Q&A + Homework**

Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

Class se poochho: Aaj ka sabse useful node kaunsa laga? · Homework mein pehla step kya hoga? · Aaj ki class 1 line mein summarize karo.

## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 20

**Scaling, Security & Workflow DevOps**

"Ek client se 50 clients tak — bina tootey."

☐ **Day 19 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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# 🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → B**
WEBHOOK_URL

**Q2 → C**
N8N_ENCRYPTION_KEY

**Q3 → D**
Docker volume

**Q4 → A**
Reverse proxy + SSL cert

### ✅ Challenge ka solution

Down? (true) → Wait 1 min → HTTP Ping n8n (copy) → IF statusCode != 200 → Down Alert.

## 📚 Official docs

- Hosting n8n ↗
- Docker install ↗
- Environment variables ↗
- Server setups ↗

## ▶ Videos

- n8n self host docker compose ↗
- n8n hostinger vps install ↗

**Doubt poochho**
WhatsApp: +91 83838 91716
wa.me/918383891716

**Day 19 lesson page**
Quiz, copy buttons, videos
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